

BOOK TWO.
THE ROOTS.

Seventh Section.
Reality of the Roots.

§ 76. Generalities on the reality of roots of equations and on the discriminant

In this section we shall occupy ourselves with the question how many roots of an algebraic equation are real. The coefficients of the equation are assumed here to be real numbers. We shall call such equations, for short, real equations, and begin with some general considerations.

The real equation of degree n

$$f(x) = 0$$

has, as we saw in the third section, n roots, which are either real or imaginary, that is, of the form $\xi + i\eta$, with real ξ, η . The number n of roots is obtained in general only when, in certain circumstances, one counts a root repeatedly, namely $(m + 1)$ times when, together with $f(x)$, the first m derivatives of $f(x)$ vanish. Since a complex expression of the form $X + iY$ is equal to zero only when the two real components X and Y vanish separately, so that with $X + iY$ also $X - iY$ vanishes, and since with real coefficients, if $f(\xi + i\eta) = X + iY$, one obtains $f(\xi - i\eta) = X - iY$, it follows from $f(\xi + i\eta) = 0$ that also $f(\xi - i\eta) = 0$. Thus to every imaginary root $\xi + i\eta$ there belongs a second imaginary root, distinct from it, $\xi - i\eta$, that is, the conjugate imaginary root. Since with the derivatives $f'(\xi + i\eta), f''(\xi + i\eta), \dots$ the conjugate quantities $f'(\xi - i\eta), f''(\xi - i\eta), \dots$ also vanish, conjugate imaginary roots have the same multiplicity.

It follows that the imaginary roots always occur in an even number, and that an equation of odd degree must always have at least one real root.

Our next aim will be to determine the number of real roots, without solving the equation, directly from the values of certain rational functions of the coefficients of the equation.

A very important role is played here by the discriminant, and we must first discuss its significance for our question.

In § 46 we defined the discriminant as the product of the squares of all differences of roots,

$$(x_1 - x_2)^2(x_1 - x_3)^2 \cdots (x_{n-1} - x_n)^2,$$

multiplied further by a_0^{2n-2} , when a_0 is the coefficient of the highest power of the unknown in the given equation. We showed there how the discriminant can be computed as a rational function of the coefficients. The factor a_0^{2n-2} is always positive and could here, without essential restriction of generality, be assumed equal to 1. As before, we shall use the letter D for the discriminant.

The discriminant vanishes if and only if two of the roots are equal.

Suppose that $f(x)$ has been freed of repeated factors by separating off the greatest common divisor of $f(x)$ and $f'(x)$, which is done by rational calculation. Then D will not vanish.

If x_1 and x_2 are real, then $(x_1 - x_2)^2$ is positive. If x_1 is real and x_2 imaginary, then there is a root x'_2 conjugate to x_2 , and the product

$$(x_1 - x_2)(x_1 - x'_2)$$

is positive as the product of two conjugate imaginary quantities, and therefore its square is positive as well.

If x_1 and x_2 are both imaginary but not conjugate, then the product

$$(x_1 - x_2)(x'_1 - x'_2)$$

and its square are positive.

Finally, if x_1 and x_2 are conjugate imaginary roots, then their difference is purely imaginary and its square is negative.

Thus in the product by which we have defined the discriminant there occur exactly as many negative factors as there are pairs of conjugate imaginary roots. We conclude from this the important fundamental theorem:

If $f(x) = 0$ is a real equation without repeated roots, then the discriminant is positive or negative according as the number of pairs of conjugate imaginary roots is even or odd.

The case in which only real roots are present belongs to those in which the discriminant is positive.

For quadratic and cubic equations, in which only one pair of imaginary roots can occur, this main theorem already completes the distinction of the possible cases with respect to reality of the roots, that is, the determination.

§ 77. Discussion of the quadratic and cubic equations

For the quadratic equation

$$a_0x^2 + a_1x + a_2 = 0$$

we have (§ 46)

$$D = a_1^2 - 4a_0a_2 \begin{cases} > 0, & \text{real roots,} \\ < 0, & \text{imaginary roots.} \end{cases}$$

For the cubic equation

$$a_0x^3 + a_1x^2 + a_2x + a_3 = 0$$

one has

$$D = a_1^2a_2^2 + 18a_0a_1a_2a_3 - 4a_0a_2^3 - 4a_1^3a_3 - 27a_0^2a_3^2,$$

and

$$\begin{aligned} D &> 0, & \text{three real roots,} \\ D &< 0, & \text{one real and two imaginary roots.} \end{aligned}$$

The case $D = 0$ also gives no further distinctions here; for then the two roots of the quadratic equation are equal and real, and of the three roots of the cubic equation two are equal and all three are real. Indeed, an imaginary root cannot occur here twice, since then the conjugate root would also occur twice; nor can the single root be imaginary, since then a second would have to be present.

For the cubic equation it remains only to find the condition that all three roots be equal.

In that case the left hand side of the cubic equation must be a complete cube, hence

$$a_0x^3 + a_1x^2 + a_2x + a_3 = a_0(x - \alpha)^3.$$

Comparison of the two sides of this identity gives

$$a_1 = -3a_0\alpha, \quad a_2 = 3a_0\alpha^2, \quad a_3 = -a_0\alpha^3,$$

from which, by eliminating α , one obtains

$$a_1^2 - 3a_0a_2 = 0, \quad a_1a_2 - 9a_0a_3 = 0. \quad (1)$$

As a consequence, multiplying the first of these equations by a_2 and the second by a_1 and subtracting, one obtains

$$a_2^2 - 3a_1a_3 = 0;$$

and if these conditions are fulfilled, then conversely

$$f(x) = a_0 \left(x + \frac{a_1}{3a_0} \right)^3,$$

which expresses the equality of all three roots.

For the cubic equation

$$x^3 + a_1x^2 + a_2x + a_3 = 0$$

we can decide completely not only on the reality but also on the signs of the roots.

Let α, β, γ be the three roots. Then

$$a_1 = -(\alpha + \beta + \gamma), \quad a_2 = \alpha\beta + \alpha\gamma + \beta\gamma, \quad a_3 = -\alpha\beta\gamma. \quad (2)$$

If now $D < 0$, so that only one root, say α , is real, then $\beta\gamma$ is positive and α is negative or positive according as a_3 is positive or negative. If, however, D is positive, so that all three roots are real, then if α, β, γ are positive, (2) gives

$$a_1 < 0, \quad a_2 > 0, \quad a_3 < 0. \quad (3)$$

These conditions are necessary for all three roots to be positive. They are also sufficient. For if one or three roots are negative, then $a_3 > 0$. But if two roots, say β, γ , are negative, then either

$$a_1 \geq 0 \quad \text{or} \quad \alpha > -(\beta + \gamma).$$

In the latter case one has

$$a_2 < \beta\gamma - (\beta + \gamma)^2 = -\beta^2 - \beta\gamma - \gamma^2,$$

so that a_2 is negative.

If finally one root is equal to 0, then necessarily a_3 vanishes.

Replacing x by $-x$, one concludes that for three negative roots the necessary and sufficient condition is

$$a_1 > 0, \quad a_2 > 0, \quad a_3 > 0. \quad (4)$$

Under the assumption of a positive discriminant we therefore obtain the following table:

a_1	a_2	a_3	number of positive roots
–	+	–	3
–	–	–	1
+	+	–	1
+	–	–	1
+	+	+	0
–	–	+	2
–	+	+	2
+	–	+	2

where the last column gives the number of positive roots. We can state the result of this consideration as follows:

For positive discriminant, the number of positive roots of the cubic equation is equal to the number of variations of sign in the sequence

$$1, a_1, a_2, a_3,$$

where by a variation of sign we mean the succession of a positive and a negative quantity, or of a negative and a positive quantity.

If one of the two quantities a_1, a_2 vanishes, then, since not all roots can have the same sign, we have one or two positive roots.

If $a_3 = 0$, then the discriminant reduces to

$$a_2^2(a_1^2 - 4a_2),$$

and if this is positive, the cubic equation has one vanishing root and two further real roots. These are positive when

$$a_1 < 0, \quad a_2 > 0,$$

negative when

$$a_1 > 0, \quad a_2 > 0,$$

and one of them is positive and the other negative when

$$a_1 \gtrless 0, \quad a_2 < 0.$$

§ 78. Discussion of the biquadratic equation

For equations of the fourth and fifth degrees there are either no, two, or four imaginary roots. If the discriminant is negative, there are two imaginary roots. With positive discriminant, either four or no imaginary roots may be present. To distinguish these two cases will be our next task. But first we turn to an elementary consideration of the biquadratic equation, which for simplicity we take in the shortened form

$$x^4 + ax^2 + bx + c = 0, \tag{1}$$

from which we can easily return to the general form.

Let the roots be denoted by $\alpha, \beta, \gamma, \delta$, and put, as in § 47,

$$\begin{aligned} \alpha - \beta &= v + w, & \gamma - \delta &= v - w, \\ \alpha - \gamma &= w + u, & \delta - \beta &= w - u, \\ \alpha - \delta &= u + v, & \beta - \gamma &= u - v. \end{aligned} \tag{2}$$

Then u^2, v^2, w^2 are the roots of the cubic resolvent

$$y^3 + 2ay^2 + (a^2 - 4c)y - b^2 = 0, \tag{3}$$

and the discriminant D of this cubic equation, determined by

$$27D = 4(a^2 + 12c)^3 - (2a^3 - 72ac + 27b^2)^2, \tag{4}$$

is at the same time the discriminant of the biquadratic equation (1), and the signs of u, v, w are constrained by

$$uvw = -b \tag{5}$$

[§ 37, (3)].

If the discriminant is negative, then equation (3), like (1), has two conjugate imaginary roots.

If D is positive and all four roots $\alpha, \beta, \gamma, \delta$ are real, then u, v, w are also real, and therefore their squares, the roots of (3), are positive.

If all four roots are imaginary, for instance α conjugate to β and γ conjugate to δ , it follows from (2) that v and w are purely imaginary and u is real. In this case equation (3) therefore has one positive and two negative roots.

In both cases, however, if b vanishes, one of u, v, w may also be equal to zero.

Taking account of the results of the preceding paragraph on the cubic equation, we arrive at the following result:

The necessary and sufficient condition for the existence of four distinct real roots is

$$D > 0, \quad a < 0, \quad a^2 - 4c > 0. \tag{6}$$

In all other cases in which D is positive, the equation has four imaginary roots.

We can also carry out the discussion completely for $D = 0$, that is, in the case of equality of two roots.

Suppose $\gamma = \delta$. Since we have assumed $\alpha + \beta + \gamma + \delta = 0$, (2) gives

$$2v = 2w = \alpha - \beta, \quad 2u = \alpha + \beta - 2\gamma = 2(\alpha + \beta),$$

and hence, from (3),

$$\begin{aligned} -4a &= 2(\alpha + \beta)^2 + (\alpha - \beta)^2, \\ 16(a^2 - 4c) &= (\alpha - \beta)^2[8(\alpha + \beta)^2 + (\alpha - \beta)^2]. \end{aligned} \quad (7)$$

First consider the case where also $\alpha = \beta$. From (7) we obtain the condition $a^2 - 4c = 0$, and the first equation (7) shows that a is negative when α and γ are real, and positive when α and γ are conjugate (and then, because $\alpha + \gamma = 0$, purely imaginary). Therefore:

The biquadratic equation has two pairs of equal real roots when

$$D = 0, \quad a < 0, \quad a^2 - 4c = 0, \quad (8)$$

and two pairs of equal imaginary roots when

$$D = 0, \quad a > 0, \quad a^2 - 4c = 0. \quad (9)$$

If, however, α is different from β , then γ must be real, and (7) shows that if α and β are real, a is negative and $a^2 - 4c$ positive, whereas if α and β are conjugate imaginary, either a is positive or $a^2 - 4c$ is negative. For in this case $(\alpha - \beta)^2$ is negative, and if $2(\alpha + \beta)^2 + (\alpha - \beta)^2$ is positive, then certainly $8(\alpha + \beta)^2 + (\alpha - \beta)^2$ must also be positive.

Thus we can supplement (6) by saying:

The necessary and sufficient condition for the existence of four real roots, of which two, but not more, may be equal, is

$$D \geq 0, \quad a < 0, \quad a^2 - 4c > 0. \quad (10)$$

The necessary and sufficient condition for three equal roots, which are necessarily all real, is by (2)

$$u^2 = v^2 = w^2.$$

Thus the cubic resolvent (3) must have three equal roots, and for this the necessary and sufficient conditions, by §77 (1), are

$$a^2 + 12c = 0, \quad 2a^3 - 8ac + 9b^2 = 0. \quad (11)$$

By §47, (15), (16), the two invariants A and B of the biquadratic equation are

$$A = a^2 + 12c, \quad B = 2a^3 - 72ac + 27b^2.$$

Thus equations (11) can also be written

$$A = 0, \quad B + 4aA = 0,$$

so that (11) is equivalent to

$$A = 0, \quad B = 0.$$

By eliminating c from (11), one may also derive

$$8a^3 + 27b^2 = 0. \quad (12)$$

Finally, the condition for equality of all four roots is

$$a = 0, \quad b = 0, \quad c = 0. \quad (13)$$

These relations can be made quite vivid by a geometric interpretation. Although geometric considerations do not properly belong to our plan, we shall not omit occasional indications of them for the reader.

If a, b, c are interpreted as rectangular coordinates in space, then every point of space may be regarded as the carrier of a certain biquadratic equation of the form (1), $f(x) = 0$. All equations for which a fixed number x is a root are represented by points of a plane $[f(x) = 0]$. The equation $D = 0$ is the equation of a curved surface of the fifth degree, the discriminant surface, generated by the lines of intersection of the planes $f(x) = 0$, $f'(x) = 0$; it is therefore a developable surface. It is the envelope of all planes $f(x) = 0$.

The surface has a cuspidal edge consisting of two branches, represented by equations (11), (12), and a double line determined by $b = 0$, $a^2 - 4c = 0$, hence of parabolic shape. On the side of negative a the surface intersects itself in this parabola. On the side of positive a the parabola continues as an isolated line.

The discriminant surface divides all space into three compartments, of which two are connected only along the double parabola, and these compartments contain the points corresponding to no, two, and four real roots. Let us temporarily call these three compartments $[0]$, $[2]$, $[4]$. Then $[0]$ borders $[4]$ only along the double parabola, while $[0]$ and $[4]$ each border $[2]$ along parts of the surface. We shall denote by $[0, 2]$ and $[4, 2]$ the boundary surfaces of $[0]$, $[2]$, and of $[0]$, $[4]$. The isolated part of the parabola continues into the interior of $[0]$. The cuspidal edges lie on the part of the surface that separates $[4]$ from $[2]$.

At the origin of coordinates the three regions of space and their boundary surfaces and boundary lines meet. The points of the surface parts $[0, 2]$ represent equations with two equal and two imaginary roots; the points of $[2, 4]$ represent equations with two equal and two distinct real roots.

On the double parabola there are twice two equal roots, namely real roots on the part where $[0]$ borders $[4]$, and imaginary roots on the isolated part.

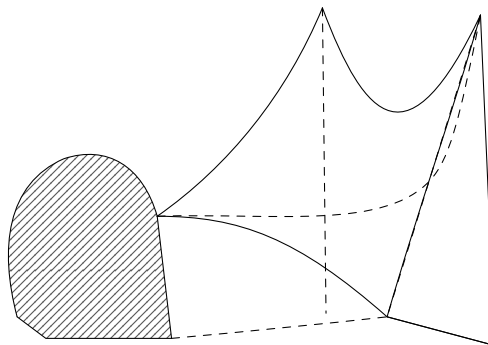


Fig. 5.

The points of the cuspidal edges represent equations with three equal roots, and the origin of coordinates represents the equation with four equal roots.

Kronecker was the first to point out the discriminant surface and its significance for the discussion of the biquadratic equation. A model of the surface was constructed by Kerschensteiner.

The analytic expression for the conditions for reality of the roots can be put into a form in which only the covariants of the biquadratic form occur.

The starting point is given by the expressions for the functions ψ_1, ψ_2, ψ_3 given in § 66, (6). If in those expressions we interchange α with β , then ψ_1, ψ_2, ψ_3 pass into $\psi_1, -\psi_3, -\psi_2$; and if at the same time we interchange α with β and γ with δ , then ψ_1, ψ_2, ψ_3 pass into $\psi_1, -\psi_2, -\psi_3$. From this the following results.

Put any real value for the variable x . If the roots $\alpha, \beta, \gamma, \delta$ of the biquadratic form are all real, then ψ_1, ψ_2, ψ_3 are also real.

If α, β are conjugate imaginary and γ, δ real, then interchanging α and β is equivalent to replacing i by $-i$. Thus in this case ψ_1 is real, while ψ_2, ψ_3 are conjugate imaginary.

Finally, if α, β and γ, δ are two pairs of conjugate imaginary roots, then α is interchanged with β and γ with δ when i passes into $-i$; therefore in this case ψ_1 is real and ψ_2, ψ_3 are purely imaginary.

If the four roots are real, then $\psi_1^2, \psi_2^2, \psi_3^2$ are real and positive. If two roots are real, then ψ_1^2 is real and positive, while ψ_2^2, ψ_3^2 are conjugate imaginary. If four roots are imaginary, then ψ_1^2 is positive, while ψ_2^2, ψ_3^2 are real and negative.

Now in § 66, (7) we set up the cubic equation whose roots are $\psi_1^2, \psi_2^2, \psi_3^2$, and also formed its discriminant. In the preceding paragraph we learned a criterion for the number of positive roots of a cubic equation, and from it the following result is obtained:

If $D < 0$, then the biquadratic equation has two real and two imaginary roots.

If $D > 0$, then in order that four real roots should be present there must be three variations of sign in the sequence

$$1, \quad H, \quad H^2 - 16Af^2, \quad -I^2,$$

that is, one must have

$$H < 0, \quad H^2 - 16Af^2 > 0.$$

In the three other possible sign combinations all four roots are imaginary. Here an arbitrary real value may be put for x .

§ 79. The Bezoutiant and its significance for reality of roots

For the investigation of the reality of the roots of an arbitrary real equation in more general cases, one may use with advantage the function which in § 73 we called the Bezoutiant of the equation.

Let

$$f(x) = x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_n = 0 \quad (1)$$

be any real equation of degree n . The Bezoutiant was defined as follows. Put

$$y = t_{n-1}f_0(x) + t_{n-2}f_1(x) + \cdots + t_0f_{n-1}(x), \quad (2)$$

where $t_0, t_1, \dots, t_{n-1}$ denote indeterminate variables and

$$f_0(x) = 1, \quad f_1(x) = x + a_1, \quad f_2(x) = x^2 + a_1x + a_2, \quad \dots$$

Then [§ 73, (10)]

$$S(y^2) = \frac{1}{n}[S(y)]^2 + B, \quad (3)$$

where B is a quadratic form in the $n - 1$ variables $t_0, t_1, \dots, t_{n-2}$, with coefficients composed rationally from the coefficients a_i . This function B we have called the Bezoutiant, and have actually formed it in the cases $n = 4$ and $n = 5$.

In the sum on the left hand side of (3),

$$y_1^2 + y_2^2 + \cdots + y_n^2, \quad (4)$$

the $y_1, y_2, \dots, y_n$ are linear functions of the variables $t_0, t_1, \dots, t_{n-1}$; hence in (3) we have a transformation of the quadratic form

$$\frac{1}{n}[S(y)]^2 + B = \Phi(t_0, t_1, \dots, t_{n-1}) \quad (5)$$

into a sum of n squares.

For the moment we make no restrictive assumptions about equality or distinctness of the roots of $f(x)$. We must first examine to what extent the linear functions $y_1, y_2, \dots, y_n$ are independent. In this respect the following theorem holds.

If $x_1, x_2, \dots, x_\nu$ are distinct roots of equation (1), then $y_1, y_2, \dots, y_\nu$ are linearly independent.

For suppose that there exists among $y_1, y_2, \dots, y_\nu$ a linear relation, identical with respect to t ,

$$\alpha_1 y_1 + \alpha_2 y_2 + \cdots + \alpha_\nu y_\nu = 0,$$

whose coefficients α are independent of the t and are not all zero. This would split into the following equations:

$$\begin{aligned}\alpha_1 f_0(x_1) + \alpha_2 f_0(x_2) + \cdots + \alpha_\nu f_0(x_\nu) &= 0, \\ \alpha_1 f_1(x_1) + \alpha_2 f_1(x_2) + \cdots + \alpha_\nu f_1(x_\nu) &= 0, \\ &\vdots \\ \alpha_1 f_{\nu-1}(x_1) + \alpha_2 f_{\nu-1}(x_2) + \cdots + \alpha_\nu f_{\nu-1}(x_\nu) &= 0.\end{aligned}$$

Since $\nu \leq n$, the determinant of the first ν of these equations would have to vanish, so that

$$\begin{vmatrix} f_0(x_1) & f_1(x_1) & \cdots & f_{\nu-1}(x_1) \\ f_0(x_2) & f_1(x_2) & \cdots & f_{\nu-1}(x_2) \\ \vdots & \vdots & & \vdots \\ f_0(x_\nu) & f_1(x_\nu) & \cdots & f_{\nu-1}(x_\nu) \end{vmatrix} = 0,$$

or

$$\begin{vmatrix} 1 & x_1 + a_1 & x_1^2 + a_1 x_1 + a_2 & \cdots \\ 1 & x_2 + a_1 & x_2^2 + a_1 x_2 + a_2 & \cdots \\ \vdots & \vdots & \vdots & \\ 1 & x_\nu + a_1 & x_\nu^2 + a_1 x_\nu + a_2 & \cdots \end{vmatrix} = 0.$$

By repeated use of the theorem on addition of columns, § 22, (VII), this determinant can be brought to the form

$$\begin{vmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{\nu-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{\nu-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_\nu & x_\nu^2 & \cdots & x_\nu^{\nu-1} \end{vmatrix} = \prod_{1 \leq r < s \leq \nu} (x_r - x_s),$$

and hence it cannot vanish when, as assumed, $x_1, x_2, \dots, x_\nu$ are distinct.

When, therefore, in the sum (4) all mutually equal terms are collected, there remain as many squares of linearly independent functions as equation (1) has distinct roots.

If x_1 is a real root of (1), then y_1^2 is a positive square in the sum (4). But if x_1 and x_2 are conjugate imaginary, then also y_1 and y_2 split into two conjugate imaginary components

$$y_1 = u + vi, \quad y_2 = u - vi,$$

where u and v are real linear functions of the t . Therefore

$$y_1^2 + y_2^2 = 2u^2 - 2v^2.$$

Thus by (4) the function $\Phi(t)$ defined by (5) is decomposed into a sum of squares whose number equals the number of distinct roots of $f(x) = 0$, and among which there are as many negative squares as there are pairs of imaginary roots among these distinct roots.

By the law of inertia for quadratic forms (§ 58), these numbers remain the same under any other real linear transformation of the quadratic form into a sum of squares. Now $S(y)$ is a real linear function of the t , and B no longer contains the variable t_{n-1} . Therefore formula (5) gives the following theorem:

I. If the Bezoutiant B is decomposed by a real linear transformation into a sum of π positive and ν negative squares, not reducible to a smaller number, then $\pi + \nu + 1$ is the number of distinct roots of $f(x) = 0$; and ν is the number of pairs of conjugate imaginary roots among them, while $\pi - \nu + 1$ is the number of real roots.

If the determinant of B , and hence also the discriminant of $f(x)$ (§ 73), is non-zero, then $\pi + \nu + 1 = n$, and ν is less than or at most equal to $\frac{1}{2}n$.

Thus the Bezoutiant is a quadratic form of a special nature, expressed precisely by the fact that among the squares into which it can be decomposed at most $\frac{1}{2}n$ can be negative.

§ 80. The inertia of forms of the second degree

By the theorem of the preceding paragraph we have been led to the investigation of homogeneous functions of the second degree with real coefficients. For these functions we encountered in § 58, under the name of the law of inertia, a theorem which forms the basis of what follows. The theorem asserted that however one transforms a quadratic form in m variables into a sum of positive and negative squares of linearly independent linear functions, and this can be done in infinitely many different ways, the number of positive and likewise the number of negative squares is always the same.

Let π denote the number of positive and ν the number of negative squares. Then $\pi + \nu$ is at most equal to m .

If we denote the difference $m - \pi - \nu$ by ρ , then, when the general quadratic form in m variables is represented as a sum of m squares, ρ may be regarded as the number of squares with vanishing coefficients.

The three numbers π, ν, ρ , none of which can be negative and whose sum equals the number of variables,

$$\pi + \nu + \rho = m, \quad (1)$$

are therefore invariant for a given quadratic form.

We must seek means of determining the numbers π, ν, ρ from the coefficients of the quadratic form. Applying these means to the Bezoutiant, we obtain criteria for inferring the number of real roots of an equation from its coefficients.

Thus let, as in § 56,

$$\varphi(x_1, x_2, \dots, x_m) = \sum_1^m a_{i,k} x_i x_k, \quad a_{i,k} = a_{k,i}, \quad (2)$$

be a quadratic form in m variables with real coefficients $a_{i,k}$.

The determinant

$$R = \sum \pm a_{1,1} a_{2,2} \cdots a_{m,m} \quad (3)$$

indicates, by its vanishing, that φ can be transformed by a linear transformation into a function of fewer than m variables, so that ρ is greater than zero.

The determinant R is symmetric. Among its subdeterminants of various orders there occur certain ones which are again symmetric determinants, namely those obtained from R by deleting rows and columns that meet in diagonal terms. Thus, if $\alpha, \beta, \gamma, \dots$ denote any of the indices $1, 2, \dots, m$, they are denoted by

$$\sum \pm a_{\alpha,\alpha} a_{\beta,\beta} a_{\gamma,\gamma} \cdots. \quad (4)$$

We shall call these the principal subdeterminants of R .

We shall determine the number of principal subdeterminants.

The number of μ -rowed principal subdeterminants is equal to the number of ways of selecting from the sequence of indices $1, 2, 3, \dots, m$ groups of μ distinct indices, hence equal to the number of combinations of m elements taken μ at a time; this number is the binomial coefficient B_μ^m . Thus the number of all principal subdeterminants, if we count the determinant R itself and also the unit as a zero-row determinant, is

$$1 + m + \frac{m(m-1)}{1 \cdot 2} + \cdots + m + 1 = 2^m.$$

But usually only a small part of these is actually to be considered.

The indices $1, 2, 3, \dots, m$ can be arranged in $\Pi(m) = 1 \cdot 2 \cdot 3 \cdots m$ different ways. If from any one of these arrangements we form the determinant

$$\begin{vmatrix} a_{1,1} & a_{1,2} & a_{1,3} & \cdots & a_{1,m} \\ a_{2,1} & a_{2,2} & a_{2,3} & \cdots & a_{2,m} \\ a_{3,1} & a_{3,2} & a_{3,3} & \cdots & a_{3,m} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m,1} & a_{m,2} & a_{m,3} & \cdots & a_{m,m} \end{vmatrix},$$

then a sequence of principal subdeterminants can be derived from it, as indicated by the strokes: each preceding determinant is obtained from the following one by removing the last row and column. Thus

$$R_0 = 1, \quad R_1 = a_{1,1}, \quad R_2 = a_{1,1}a_{2,2} - a_{1,2}^2, \\ R_3 = \begin{vmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{vmatrix}, \quad \dots, \quad R_m = R.$$

Such a system, when arranged in descending order

$$R_m, R_{m-1}, \dots, R_1, R_0,$$

will be called a chain of principal subdeterminants. There are $\Pi(m)$ such chains, in all of which the first term is R and the last term is 1.

§ 81. Quadratic forms with vanishing determinant

If the determinant R of the quadratic form $\varphi(x_1, x_2, \dots, x_m)$ defined in (3), § 80, vanishes, then, as we already saw in § 57, φ can be expressed by fewer than m independent linear functions of x . If we denote by $m - \rho$ the least number of variables through which φ can be expressed, then ρ has the same meaning as in the preceding paragraph.

If among the principal subdeterminants of R there is a k -rowed one which is not zero, say

$$R_k = \sum \pm a_{1,1}a_{2,2} \cdots a_{k,k},$$

then, putting

$$x_{k+1} = 0, \quad x_{k+2} = 0, \quad \dots \quad x_m = 0,$$

one obtains

$$\varphi(x_1, x_2, \dots, x_k, 0, 0, 0),$$

a quadratic form in the k variables $x_1, x_2, \dots, x_k$ whose determinant R_k is non-zero, and which therefore cannot be expressed by fewer than k variables. Still less can the function φ in the unrestricted variables $x_1, x_2, \dots, x_m$ depend on fewer than k variables. Hence

$$\rho \leq m - k. \tag{1}$$

Suppose now that the form $\varphi(x_1, x_2, \dots, x_m)$ can be expressed by k mutually independent linear forms in x , which we denote by

$$\begin{aligned} \xi_1 &= \beta_1^{(1)} x_1 + \beta_2^{(1)} x_2 + \cdots + \beta_k^{(1)} x_k + \beta_{k+1}^{(1)} x_{k+1} + \cdots + \beta_m^{(1)} x_m, \\ &\dots \\ \xi_k &= \beta_1^{(k)} x_1 + \beta_2^{(k)} x_2 + \cdots + \beta_k^{(k)} x_k + \beta_{k+1}^{(k)} x_{k+1} + \cdots + \beta_m^{(k)} x_m. \end{aligned} \tag{2}$$

Since $\xi_1, \xi_2, \dots, \xi_k$ are to be linearly independent, at least one of the k -rowed determinants formed from the matrix

$$\begin{array}{cccc} \beta_1^{(1)} & \beta_2^{(1)} & \cdots & \beta_m^{(1)} \\ \vdots & \vdots & & \vdots \\ \beta_1^{(k)} & \beta_2^{(k)} & \cdots & \beta_m^{(k)} \end{array}$$

must be non-zero. For if all of them were zero, the unknowns $h_1, h_2, \dots, h_k$, not all of which are to vanish, could be determined from the equations

$$h_1\beta_s^{(1)} + h_2\beta_s^{(2)} + \cdots + h_k\beta_s^{(k)} = 0, \quad s = 1, 2, 3, \dots, m$$

(§24), and the $\xi_1, \xi_2, \dots, \xi_k$ would satisfy an equation

$$h_1\xi_1 + h_2\xi_2 + \cdots + h_k\xi_k = 0,$$

and hence would not be independent.

We may therefore, without restricting generality, assume that the determinant

$$\sum \pm \beta_1^{(1)} \beta_2^{(2)} \cdots \beta_k^{(k)}$$

is non-zero; then the equations (2) can be solved for the variables $x_1, x_2, \dots, x_k$.

We give these solutions the form

$$\begin{aligned} x_1 &= y_1 + \alpha_{k+1}^{(1)} x_{k+1} + \cdots + \alpha_m^{(1)} x_m, \\ x_2 &= y_2 + \alpha_{k+1}^{(2)} x_{k+1} + \cdots + \alpha_m^{(2)} x_m, \\ &\dots \\ x_k &= y_k + \alpha_{k+1}^{(k)} x_{k+1} + \cdots + \alpha_m^{(k)} x_m, \end{aligned} \tag{3}$$

where the y_s are homogeneous linear combinations of the ξ_s , determined by the equations

$$\begin{aligned} \beta_1^{(1)} y_1 + \beta_2^{(1)} y_2 + \cdots + \beta_k^{(1)} y_k &= \xi_1, \\ \beta_1^{(2)} y_1 + \beta_2^{(2)} y_2 + \cdots + \beta_k^{(2)} y_k &= \xi_2, \\ &\dots \\ \beta_1^{(k)} y_1 + \beta_2^{(k)} y_2 + \cdots + \beta_k^{(k)} y_k &= \xi_k. \end{aligned}$$

The α are new coefficients, derived in an easily surveyable way from the β , but their formation is of no further importance here.

By hypothesis, the function $\varphi(x_1, x_2, \dots, x_m)$ can be expressed by $\xi_1, \xi_2, \dots, \xi_k$, and therefore also by $y_1, y_2, \dots, y_k$ alone. Denote this expression by $\Phi(y_1, y_2, \dots, y_k)$. Then, by means of (3), we have the identity

$$\varphi(x_1, x_2, \dots, x_k, x_{k+1}, \dots, x_m) = \Phi(y_1, y_2, \dots, y_k), \tag{4}$$

and if in it we put $x_{k+1}, x_{k+2}, \dots, x_m = 0$, then

$$\Phi(x_1, x_2, \dots, x_k) = \varphi(x_1, x_2, \dots, x_k, 0, 0, \dots, 0), \tag{5}$$

which, since the names of the variables are immaterial, determines Φ completely. We may therefore write

$$\begin{aligned} \Phi(y_1, y_2, \dots, y_k) &= \varphi(y_1, y_2, \dots, y_k, 0, 0, \dots, 0) \\ &= \varphi(x_1, x_2, \dots, x_m)'. \end{aligned} \tag{6}$$

Thus Φ is obtained from φ by setting the last $m - k$ variables (for the ordering chosen here) equal to zero:

$$\Phi(x_1, \dots, x_k) = \sum_1^k a_{r,s} x_r x_s. \tag{7}$$

The determinant of Φ is one of the principal subdeterminants of R ; it is obtained by deleting from R the last $m - k$ rows and columns.

Suppose that φ cannot be expressed by fewer than k variables, so that $k = m - \rho$. Then the determinant of this function Φ cannot vanish.

From this and from what was proved above, we obtain the theorem:

II. If all principal subdeterminants of R of more than k rows vanish, while among the principal subdeterminants of k rows at least one is non-zero, then

$$\rho = m - k.$$

For (7) shows that if $\rho = m - k$, then at least one k -rowed principal subdeterminant must be non-zero; and (1) shows that if a principal subdeterminant of more than k rows is still non-zero, then

$$\rho < m - k.$$

If π, ν, ρ are the numbers of positive, negative, and vanishing squares into which φ can be decomposed, then π, ν are the numbers of positive and negative squares into which the form Φ determined by (7) can be decomposed. Thus the determination of π, ν still needs to be carried out only for this latter function, whose determinant is non-zero.